

ASSIGNMENT-7

1) Without recursion using for loop



```
word="candy"
n=${#word}
echo "The word is: $word"
echo "Length of the word is $n"
factorial() {
    fact=1
    for((i=1;i<=$1;i++)); do
        fact=$((fact*i))
    done
    echo $fact
}
echo "Total number of arrangements of letter:"
result=$(factorial $n)
echo $result
```

Output:

```
(base) mmcoe@mmcoe-System-Product-Name:~$ vim f.sh
(base) mmcoe@mmcoe-System-Product-Name:~$ chmod +x f.sh
(base) mmcoe@mmcoe-System-Product-Name:~$ ./f.sh
The word is: candy
Length of the word is 5
Total number of arrangements of letter:
120
```

2) Recursion using if...else



```
word="candy"
n=${#word}
echo "The word is: $word"
echo "Length of the word is $n"
fact=1
factorial() {
    if [ $n -gt 1 ]; then
        fact=$((fact*n))
        n=$((n-1))
        factorial
    else
        echo "Ttoal number of arrangement of letters: $fact"
    fi
}
factorial
```

Output:

```
(base) mmcoe@mmcoe-System-Product-Name:~$ vim f.sh
(base) mmcoe@mmcoe-System-Product-Name:~$ chmod +x f.sh
(base) mmcoe@mmcoe-System-Product-Name:~$ ./f.sh
The word is: candy
Length of the word is 5
Ttoal number of arrangement of letters: 120
```

3) Without recursion using if-else

```
word="candy"
n=${#word}

echo "The word is : $word"
echo "The length is : $n"

fact=1

if [ $n -le 1 ]
then
    fact=1
else
    for((i=1;i<=$n;i++))
    do
        fact=$((fact * i))
    done
fi

echo "The final result is : $fact"
```

Output:

```
mmcoe@mmcoe-System-Product-Name:~$ vim Assign7_2.sh
mmcoe@mmcoe-System-Product-Name:~$ chmod +x Assign7_2.sh
mmcoe@mmcoe-System-Product-Name:~$ ./Assign7_2.sh
The word is : candy
The length is : 5
The final result is : 120
```